Mira Chami

Software Engineer

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Software engineer building cross-platform desktop applications, backend services and real-time systems, from architecture decisions through CI and release. Experience spans React/Rust clients, asynchronous Node.js services, LLM integration and multiplayer architecture, alongside two shipped mobile games.

PROFESSIONAL EXPERIENCE

Tencent

Nov 2023 – Present

Game Tech R&D Engineer · London, UK

Desktop AI Companion

React, Rust, Tauri

- Led the Electron-to-Tauri evaluation and migration, comparing performance, bundle size, multi-window behaviour and desktop APIs through working prototypes; recommended Tauri v2 and migrated the core features.
- Designed the **React/Rust client architecture**: React UI invokes Rust commands through Tauri; the Rust client sends prompts with session and game context over HTTP and streams responses back via SSE, with progressive rendering and persistent per-user conversation history.
- Built a multi-window application (chat, companion, avatar, settings) with shared messaging, overlays, hotkeys and tray access; integrated an external avatar service over gRPC.
- **Automated Windows/macOS** setup, builds, installers and packaging in CI, with Steam isolated behind feature flags; added structured logging and diagnostic panels to trace failures across frontend, Rust and platform layers.

AI-Backed Investigation Game

Node.js, Express, Gemini, Godot

- Built Node.js/Express services for an LLM-backed game, with request validation, caching, session restoration, timeouts, cancellation and retries.
- Enabled data-integrity and UI testing without live model calls through mocked responses. Built a versioned data pipeline with JSON Schema validation, cross-reference checks and a version-comparison viewer.

Online Services Integration

UE5, C++, Blueprints

- Integrated and validated InGame, Tencent’s public online services framework, in a UE5/C++ multiplayer demo covering authentication, lobbies, matchmaking, inventory and achievements.
- Supported partner studios by reproducing issues across plugin, backend and UI boundaries; implemented PlayStation 5 support and dedicated-server debugging; modified CI pipelines.

Co-op Game Systems

Godot, GDScript

- Designed server-authoritative networking supporting Steam P2P, LAN, listen and dedicated servers, with local multi-client testing.
- Built a validated action API for an LLM director to spawn enemies and grant item rewards, plus a Godot editor plugin for authoring and testing actions. Designers used the plugin to create behaviour timelines for the AI director. Profiled procedural generation to identify CPU-bound bottlenecks.

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PROFESSIONAL EXPERIENCE (CONTINUED)

The Video Games Studio (VGS)

Jun 2020 – Aug 2022

Game Programmer · Beirut, Lebanon

- **Shipped a mobile runner** from first build to store release, covering gameplay, tuning, monetisation and backend integration.
- Owned architecture for a published puzzle game, including persistence, progression and replayable player-created content.
- Integrated PlayFab and third-party backend services across Unity projects; built a separate application to manage competition cycles and retrieve winner data.
- Led programming for a Unity prototype, from design documents and technical planning to implementation.
- Mentored developers, ran code reviews, supported technical interviews and planned work in Jira and Trello; maintained software architecture documentation.

TECHNICAL TOOLKIT

Languages:
TypeScript · JavaScript · Rust · C++ · C# · Python · GDScript · Java · C

Frontend & Desktop:
React · Tauri · Electron · Native API Integration · Cross-Platform Desktop Development · Multi-Window UI

Backend & Networking:
Node.js · Express · Asynchronous Programming · Caching · HTTP · SSE · gRPC · Protobuf · InGame · PlayFab · Steamworks · Photon PUN 2

Build & Engineering:
CI · Build & Release Automation · Automated Testing · Structured Logging · Performance Profiling · Git · Perforce

Applied AI:
LLM Integration · Google Search Grounding · Structured Outputs (JSON Schema) · LLM Output Validation

Engines & Graphics:
Unreal Engine 5 · Unity · Godot · Blueprints · Unity URP · Three.js · PixiJS

EDUCATION

MSc Games Development (Programming) · Kingston University London · 2022 – 2023

BSc Computer Science · American University of Beirut · 2017 – 2021

BA Economics · American University of Beirut · 2016 – 2020

MISCELLANEOUS

Languages: English, French and Arabic - native